Data Analysis and Storytelling

Regression analysis - discrete spatial models

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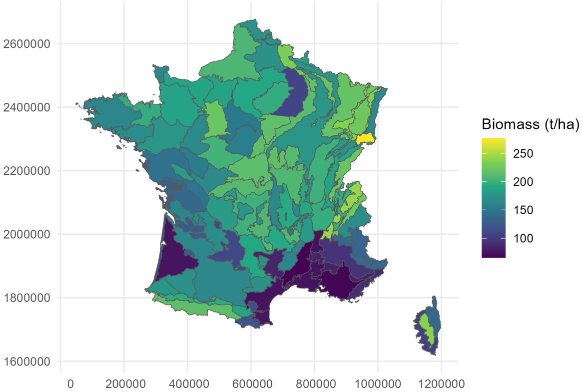
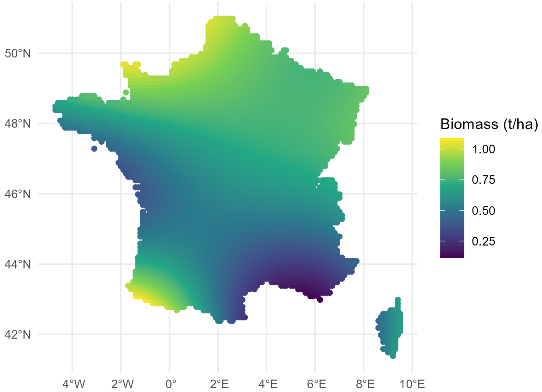
FIRS - Track 1

Two main types of spatial models



Discrete spatial units

Two main types of spatial models



Continuous spatial field

Non-spatial models for discrete spatial units

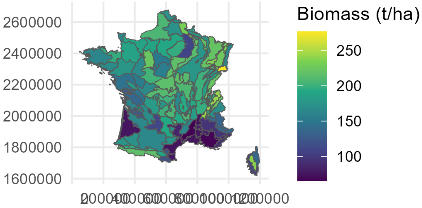
- Spatial dependency can violate the independence assumption of the observation and bias uncertainty estimation
- Just like with temporal structure, non-spatial hierarchical models can also be used to model discrete spatial units
- Such a model can be written:

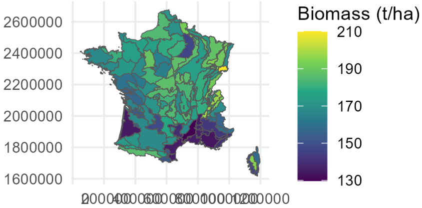
$$biomass_s \sim N(\mu_s, \sigma_{res})$$

$$log(\mu_s) = \beta_0 + \gamma_s + ...$$

$$\gamma_s \sim N(0, \sigma_{spat})$$

Non-spatial models for discrete spatial units

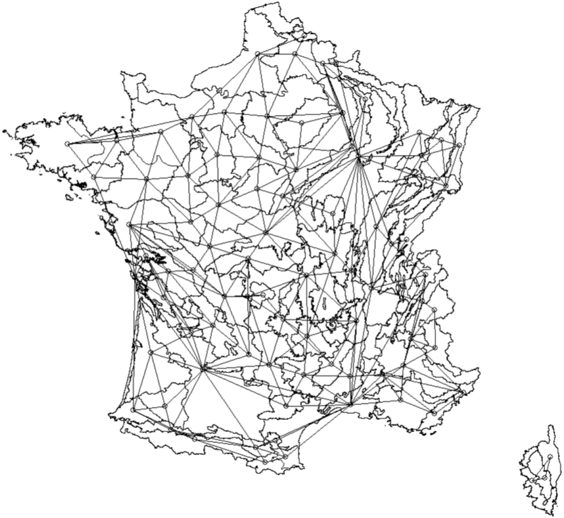




The model fits the patterns in the data pretty well. It allows to account for spatial dependency without explicit modelling of the spatial structure.

Spatial models for discrete spatial units

Explicit spatial model for discrete spatial units rely on the neighborhood structure (whom is connected to whom) between the units:



Spatial models for discrete spatial units

Several models exist to fit such data, an efficient one is the Besag-York-Mollié (BYM2):

$$biomass_i \sim N(\mu_i, \sigma_{res})$$

$$\mu_i = \beta_0 + \gamma_{s_i} + \dots$$

$$\gamma_{s_i} = \sigma_{spat}(\sqrt{1-\rho}\nu_i + \sqrt{p}u_i)$$

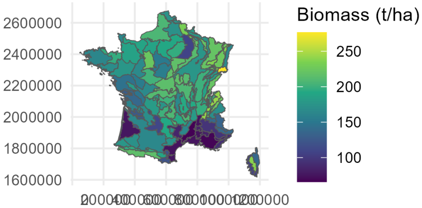
ν_i: spatially unstructured random white noise
u_i: structured spatial signal
ρ: share of spatial variation due to the spatial structure, closer to 1 stronger importance of spatially structured effect
σ_{spat}: amount of variation between the spatial units

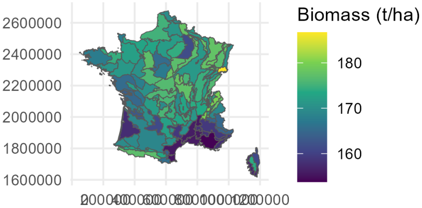
Spatial models for discrete spatial units

The value of a given spatial unit, say u_1, is the average of its connected neighbours:

$$u_i | u_{j, j \neq i, j \sim i} \sim N\left(\frac{\sum_{j=1}^{n} u_j}{n}, 1\right)$$

Spatial models for discrete spatial units





The model estimate ρ at 0.46, meaning that around half of the spatial variaiton in biomass is due to the spatial structure.

Wrap up

- Accounting for spatial dependency for discrete spatial units can be done with a varying intercept model
- Such approach is lightweight but ignore the spatial structure in the data
- All spatial models for discrete spatial units rely on neighborhood matrix which sets the connection between the units
- The BYM2 model is an efficient spatial model for discrete spatial structure it has two parameters:
  1. ρ: the amount of spatial variation due to the spatial structure
  2. σ: the amount of spatial variation